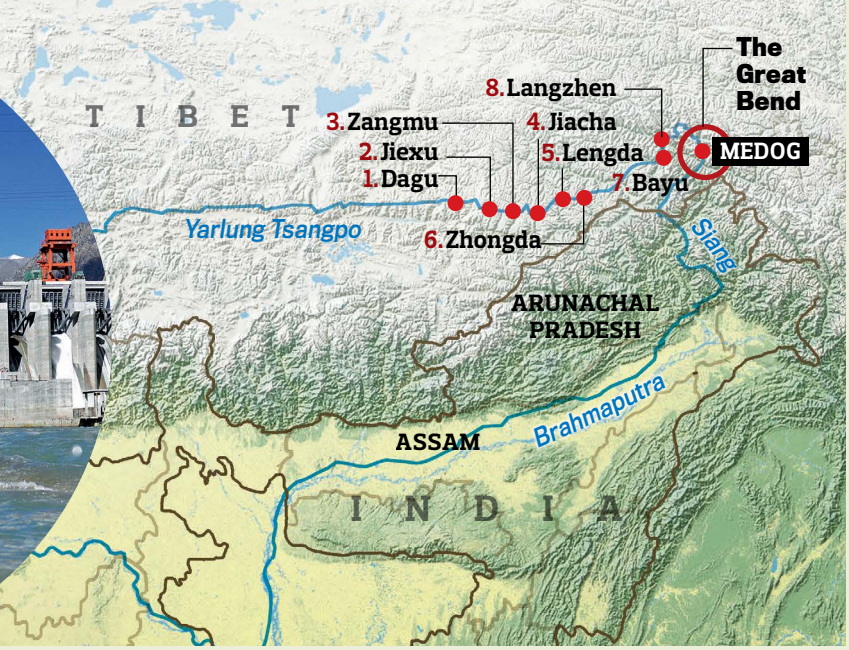


CHINA'S HYDROPOWER UNIVERSE

Besides the colossal Medog project, China has completed three other projects on the Yarlung Tsangpo, has one more under construction, and plans to build four additional ones

1. DAGU
Sangri
660 MW
Operational since 2021

2. JIEXU
Sangri
510 MW
Under construction

3. ZANGMU
Gyaca County
510 MW
Operational since 2015

4. JIACHA
Jiacha
360 MW
Operational since 2020

5. LENGDA
Gyaca
320 MW
Work likely to begin this year

6. ZHONGDA
Zhongda
320 MW
Proposed

7. BAYU
Zengqi
800 MW
Proposed

8. LANGZHEN
Lang
340 MW
Proposed

DAM OR DISASTER?

Why China's Medog dam plan is dangerous for India and the Himalayan ecology

- By controlling the Brahmaputra's flow, China gains a strategic tool in broader India-China relations, especially in border negotiations
- If China releases excess water during the monsoons, India's Northeast could face catastrophic floods; restricting the flow during dry months could cripple agriculture and energy supply
- Disruptions in the Brahmaputra's flow could devastate rice, tea and jute cultivation, threatening farmers' livelihoods
- Dam construction may block sediment flow, degrading soil fertility and accelerating riverbank erosion
- Indian power plants relying on the Brahmaputra could face blackouts if China manipulates water levels
- The region's fragile geomorphology makes dam failures from landslides or glacial lake outbursts a serious concern
- Large-scale deforestation for the project could accelerate landslides and reduce soil stability
- The Himalayas are highly earthquake-prone. The massive dam could trigger reservoir-induced seismicity and destabilise the Earth's fault lines, further increasing earthquake risks
- There is apprehension that China may use the dam to divert the Brahmaputra's flow to arid regions under the South-North Water Transfer Project

Mekong offers a stark warning for India and Bangladesh. Over two decades, Beijing has constructed 12 massive dams along the river's upper reaches, disrupting natural flows and worsening environmental stress downstream. In 2019, despite above-average rainfall, China's upstream dams hoarded record amounts of water, triggering droughts in Thailand, Cambodia and Vietnam. In 2021, it slashed the Mekong's flow by 50 per cent for three weeks—citing power-line maintenance—without prior notice, leaving millions struggling for irrigation, fisheries and drinking water. These unilateral actions stoke fears that Beijing could deploy similar tactics on the Brahmaputra.

The Medog dam isn't an isolated project but part of a grand strategy. Sayanangshu Modak, a doctoral researcher at the University of Arizona's School of Geography, Development and Environment, says the Yarlung Tsangpo stretch was marked for hydropower as early as 2003. Since the 2010s, China has steadily expanded its hydropower footprint on the Brahmaputra's upper reaches. The Zangmu dam, completed in 2015, set the stage for further projects at Dagu, Jiacha and Jiexu. Satellite imagery now reveals nearly 20 dams—large and small—along the river, signalling Beijing's long-term water ambitions. "The new move appears to be the latest step by Beijing to maintain